

EAST CRAZIES EXAMPLE SOILS SCOPE OF WORK

Date: 2026-05-07

Audience: WLG demonstration package
Purpose: show that the system can produce resource-area scopes of work for a new NEPA project from a structured project intake
Example project: East Crazy Inspiration Divide land exchange
Forest: Custer Gallatin National Forest
District: Bozeman Ranger District
Resource area: Soils

This example assumes a new project is in early planning, before late-stage NEPA materials or resource analysis packages exist. The SOW is generated from the proposed action, project location, preliminary action elements, parcel/trail/access features, likely resource triggers, and standard public source requirements.

This is the capability WLG should see: the system can translate the controlling source stack into a tight specialist-report SOW. For this soils example, the stack includes NEPA, current USDA NEPA procedures, NFMA/project-consistency requirements, Forest Service handbook and manual direction, Region 1 soil management direction, the applicable Forest Plan, soil-quality monitoring methods, National BMPs, and official soil-survey data. The SOW turns those sources into fieldwork, analysis, deliverables, and acceptance criteria that should produce a defensible soils specialist report without pretending the planning package has already made a NEPA or legal sufficiency determination.

For the East Crazies example, the soils scope is driven by:

- proposed new or rerouted trail segments;
- trailhead improvements;
- stream crossings, wetland/riparian approaches, wet soils, hydric soils, erosion, and sediment delivery;
- parcel restrictions, access/easements, grazing-related improvements, and acquired-parcel soil management needs.

The consultant product is a Soils Specialist Report or, if approved by the Forest Service soils reviewer, a short issue-screening memo. Either product must support later EA drafting, Forest Plan consistency documentation, mitigation commitments, and implementation planning.

This is a planning and contracting support SOW. It is not a final agency decision, legal sufficiency review, or NEPA compliance determination.

FIELDWORK

Before mobilization, prepare a survey-trigger table that ties each proposed action element to the source that requires field verification. Do not include generalized reconnaissance as a required task. If a survey below is not triggered, record the reason in the table.

1. Soil Survey And Activity-Area Delineation

Required for each ground-disturbing action area: proposed trail construction or reroute, trailhead work, access/staging, restoration work, crossing approaches, and acquired-parcel management areas where future NFS management could disturb soil.

- Use NRCS Web Soil Survey/SSURGO, DEM slope data, NHD/NWI, wetland/riparian layers, aerial imagery, and Forest Plan soil/watershed direction to map the soil map units and activity areas before the field visit.
- Field-verify soil map unit boundaries, slope class, landform, drainage position, hydric or wet soil indicators, erosion hazard, revegetation limits, and sediment connectivity where the action could change soil condition or water quality.
- Delineate the activity area used for Forest Plan FW-STD-SOIL-01 and MON-SOIL-01. For linear trail features, define the width, length, acreage, and rationale used to translate the trail footprint into an activity area.

2. Detrimental Soil Disturbance Survey

Required where a management activity could create or add to detrimental soil conditions in an activity area. This is the core East Crazies soils survey for trail construction or reroute, trailhead work, access/staging, restoration, and any disposal or acquisition parcel work that changes soil condition.

- Use the Forest Service Soil-Disturbance Field Guide and Forest Soil Disturbance Monitoring Protocol as the field method for classifying existing soil disturbance.
- Document compaction, rutting, displacement, puddling, surface erosion, severe burning if present, organic-matter loss, and mass movement.
- Estimate percent detrimental soil disturbance by activity area, including existing disturbance. The report must show whether the proposed action would exceed, avoid, or require restoration under the Custer Gallatin Forest Plan FW-STD-SOIL-01 threshold.
- Identify where MON-SOIL-01 pre-implementation and post-implementation monitoring is needed. If the Forest Service accepts a linear-feature substitute metric, document the metric and why it remains comparable to the Forest Plan disturbance threshold.

3. CWD And Organic-Substrate Survey

Required only if the proposed action includes vegetation activity areas, burn piles, slash treatment, forest-floor removal, revegetation/restoration, or other activity that would affect coarse woody debris, litter, duff, or organic soil cover.

- If triggered, estimate percent coarse woody debris in vegetation activity areas for MON-SOIL-02 and document litter/duff, ground cover, organic matter, and restoration substrate conditions.
- For trailhead, access, or restoration work that removes organic cover but is not a vegetation activity area, record the affected area, expected organic-cover loss, and restoration requirement.
- If no vegetation activity area or organic-substrate effect is present, mark this survey as not triggered.

4. Slope, Equipment, And Unstable-Area Survey

Required where mechanized construction, ground-based equipment, temporary access, blading, scarification, pits/trenches, burn piles, or steep/unstable landforms are proposed.

- Verify sustained grades over 40 percent where ground-based equipment could be used.
- Identify high mass-wasting potential, unstable slopes, wet soils, hydric soils, sensitive soil map units, and areas where equipment exclusion or seasonal restriction is needed.
- Record locations where temporary road blading, mechanical scarification, soil pits/trenches, or burn pile restoration would trigger Forest Plan soil guidelines FW-GDL-SOIL-01 through FW-GDL-SOIL-08.

5. Waterbody Interface And BMP Survey

Required where trail, trailhead, access, crossing, drainage, or restoration work is near streams, wetlands, riparian areas, seeps, springs, swales, drainage paths, or other sediment-routing features.

- Field-check proposed stream and wetland approaches, crossing locations, drainage outlets, slope breaks, toe slopes, and hydrologic connectivity to waterbodies.
- Identify site-specific National BMP needs: drainage spacing and outlet protection, crossing hardening, sediment control, wet-weather limits, revegetation, decompaction, and construction timing restrictions.
- Record whether the Inspiration Divide Trail reroute would remove or reduce an existing stream/wetland crossing effect and what design criteria are needed for the replacement alignment.

6. Minimum Survey Record

For each triggered survey location or activity area, record:

- date, observer, weather, soil moisture, access limits, survey trigger, and source driver;
- activity-area ID, action element, soil map unit, GPS identifier, photo IDs, and field form ID;

- protocol used, measurements taken, disturbance class, percent disturbance where required, and linear-feature substitute metric if approved;
- slope, landform, drainage position, hydric/wet soil indicators, unstable-area indicators, and sediment-connectivity notes;
- BMP, design criterion, mitigation, restoration, monitoring requirement, data gap, or revisit need.

Conduct fieldwork during snow-free conditions when the mineral soil surface, organic cover, and drainage paths can be observed. Conduct a wet-season or post-storm revisit only if waterbody connectivity, runoff, erosion, or sediment delivery cannot be determined during the main survey.

ANALYSIS

1. Authority-To-Report Crosswalk

Prepare a compact authority-to-report crosswalk before drafting effects conclusions. The crosswalk must show:

- which laws, regulations, Forest Service handbooks/manuals, regional direction, Forest Plan components, BMPs, and data standards apply to soils;
- which proposed-action element triggers each source requirement;
- which field method, analysis table, map, mitigation measure, or monitoring item satisfies each requirement;
- which requirements are not triggered and why;
- which requirements remain unresolved because field data, design detail, access, or Forest Service direction is missing.

The crosswalk must use current applicable sources. Former or superseded authority text may be noted as currentness context only; it must not drive the specialist report requirements.

2. Existing Condition

Describe the soil setting for affected action areas:

- soil map units, slopes, parent materials, drainage, erosion hazard, compaction/rutting limits, revegetation limits, hydric soil indicators, and sensitive soil attributes;
- existing roads, trails, trailheads, stream crossings, wetland/riparian areas, grazing improvements, disturbed areas, and route drainage;
- existing soil disturbance and sediment connectivity;
- soil conditions on disposal and acquisition parcels where relevant to the land exchange decision;
- data gaps and confidence limits.

3. Proposed-Action Effects

Analyze no action, proposed action, preliminary alternatives if any, and any later Forest Service-provided alternative. Address:

- soil disturbance from trail construction, trail reroute, trailhead work, crossings, access, staging, restoration, and parcel-management changes;
- whether trail work near or across waterbodies could deliver sediment, and what design features would reduce that risk;
- whether a trail reroute would reduce soil or sediment effects by removing an existing stream/wetland crossing;
- whether parcel restrictions, wetland covenants, no-subdivision restrictions, or no-mineral-development restrictions would reduce foreseeable soil disturbance after conveyance;
- whether acquired parcels would create new soil-management needs for NFS management;
- whether trailhead resurfacing, parking, toilet facilities, kiosk construction, or similar improvements would require erosion, compaction, drainage, topsoil, or revegetation controls;
- whether grazing improvements, grazing use, hazardous materials, minerals, vegetation, invasive species, restoration, or carbon assumptions would create soil handling or mitigation needs.

Quantify disturbance where useful. For linear features, use length, width, crossing count, sensitive soil crossings, wetland/riparian approaches, drainage features, and representative disturbance class instead of forcing all effects into an acreage metric.

4. Forest Plan Requirements

Evaluate and document consistency with:

- FW-DC-SOIL-01, FW-DC-SOIL-02, and FW-DC-SOIL-03;
- FW-STD-SOIL-01, including the rationale for applying or screening out the standard;
- FW-GDL-SOIL-01 through FW-GDL-SOIL-08;
- related road/trail direction, especially hydrologic disconnection, mass-wasting avoidance, and hardened stream crossings;
- invasive species, revegetation, riparian, wetland, aquatic, recreation, backcountry, and recommended-wilderness components that affect soil or sediment analysis.

The report must identify which Forest Plan components apply to the proposed action and what design, mitigation, or monitoring requirements should carry forward.

5. Mitigation, BMPs, And Monitoring

Prepare a mitigation/BMP/monitoring matrix with action element, location trigger, soil concern, required measure, responsible party, timing, and verification method. Include, as applicable:

- trail drainage spacing, rolling dips, grade reversals, waterbars, cross drains, armoring, outsloping, and sediment filtering;
- hydrologic disconnection from waterbodies and riparian management zones;
- stream-crossing hardening for trail approaches, streambeds, banks, and approaches;
- avoidance or design modification for wet soils, hydric soils, steep slopes, unstable areas, high mass-wasting potential, and sensitive soil map units;
- wet-weather restrictions, erosion control, decompaction, topsoil/duff salvage, revegetation, mulch, weed-free seed, weed prevention, restoration, and post-construction photo-point monitoring.

6. Cumulative Effects And Uncertainty

Evaluate cumulative soil effects only where there is a useful cause-and-effect relationship. Consider existing and proposed routes, trailheads, unauthorized routes, route drainage, grazing, wetland or riparian disturbance, foreseeable trail or road work, private land development risk after conveyance, and vegetation/restoration actions that could affect ground cover or erosion.

Clearly identify unresolved field limitations, assumptions, and Forest Service decisions needed before soil conclusions can be used in EA drafting, decision support, collection agreements, trail design, or implementation files.

DELIVERABLES

1. Kickoff And Data Request Log

Provide kickoff notes, received and missing data, access constraints, source materials used, stop conditions, and Forest Service decisions needed before fieldwork.

Acceptance criteria:

- identifies missing GIS, trail design, parcel, wetland, route, construction-method, or Forest Plan data;
- identifies the current laws, regulations, Forest Service handbook/manual direction, Region 1 direction, Forest Plan components, BMPs, and soil-data sources that will govern the soils work;
- identifies whether the work will proceed as a standalone soils report or issue-screening memo;
- identifies field-season and access constraints.

2. Fieldwork Plan

Provide field objectives, activity-area definitions, field maps, observation methods, field forms, data conventions, schedule, safety/access plan, and coordination needs.

Acceptance criteria:

- covers the proposed trail, reroute, trailhead, crossings, wetland/riparian interfaces, parcel restrictions, and other project-specific soil trigger areas;
- explains how soil disturbance will be documented for linear features;
- identifies whether wet-season or post-storm verification may be needed.

3. Field Data Package

Provide GPS data, completed field forms, photo log, soil-check notes, soil map unit verification, wetland/riparian and sediment-connectivity notes, and a short field findings memo.

Acceptance criteria:

- every material field conclusion is traceable to location, date, observer, photo, and activity area;
- sensitive soil, hydric/wet soil, erosion, compaction, rutting, displacement, drainage, and sediment concerns are mapped or explicitly screened out;
- data are suitable for Custer Gallatin GIS and ID-team use.

4. Analysis Tables And Map Atlas

Provide:

- soil map unit and activity-area crosswalk;
- existing and expected disturbance table;
- action-element by soil concern matrix;
- stream/wetland crossing and sediment-connectivity table;
- mitigation/BMP/monitoring matrix;
- authority-to-report crosswalk;
- Forest Plan requirements matrix;
- data-gap and uncertainty table;
- project, soil-analysis area, soil map unit, slope/erosion hazard, wetland/riparian, road/trail, observation, and mitigation maps.

Acceptance criteria:

- each required fieldwork, analysis, mitigation, monitoring, and deliverable item is tied to the controlling source that created the need;
- maps include scale, north arrow, coordinate system, data date, project boundary, activity-area IDs, and source attribution;
- tables distinguish no action, proposed action, preliminary alternatives, and any later analyzed alternative;
- linear-feature effects use useful measures and do not rely on misleading acreage-only metrics.

5. Draft Soils Specialist Report Or Issue-Screening Memo

Provide a draft report or approved issue-screening memo with affected environment, methods, spatial and temporal scale, resource indicators, direct and indirect effects, cumulative effects, mitigation/BMPs/monitoring, Forest Plan requirements, references, and appendices.

Acceptance criteria:

- directly addresses the proposed action and each soil-triggering action element;
- states whether FW-STD-SOIL-01 applies or is screened out for trail work;
- ties conclusions to field evidence, GIS evidence, current laws and regulations, Forest Service handbook/manual direction, regional direction, Forest Plan components, BMPs, official data sources, or specialist judgment.

6. EA-Ready Soils Text

Provide concise text for the future EA or appendix covering affected environment, environmental consequences, mitigation/design criteria, Forest Plan requirements, and cumulative effects if needed.

Acceptance criteria:

- suitable for public disclosure;
- references the soils report or issue-screening memo instead of duplicating all technical detail;
- identifies design criteria or monitoring that must carry into later decision or implementation files.

7. Final Report And Data Package

Provide final report or memo, final PDF, editable source file, comment-response log, final tables, map atlas, GIS layers or geodatabase/geopackage, photo log, field forms, source data inventory, metadata/readme, and data limitations memo.

Acceptance criteria:

- resolves Forest Service and ID-team comments;
- preserves source citations and field-data traceability;
- preserves the authority-to-report crosswalk so the final specialist report remains defensible;
- documents residual uncertainty and remaining Forest Service decisions;
- final PDF starts with a valid '%PDF-' header.

SOURCES CITED

- National Environmental Policy Act, 42 U.S.C. 4321 et seq.
- National Forest Management Act, 16 U.S.C. 1600 et seq.
- USDA NEPA procedures, 7 CFR part 1b:
<https://www.ecfr.gov/current/title-7/subtitle-A/part-1b>
- National Forest System land management planning, 36 CFR part 219:
<https://www.ecfr.gov/current/title-36/chapter-II/part-219>
- Forest Service Handbook (FSH) 1909.15, National Environmental Policy Act Handbook.
- USDA Forest Service Manual, Region 1 Supplement, FSM 2550 Soil Management.
- Custer Gallatin National Forest Land Management Plan:
<https://www.fs.usda.gov/r01/custergallatin/planning/forest-plan/custer-gallatin-land-management-plan-forest-plan-revision>
- Forest Service Soil Quality Monitoring resources:
<https://www.fs.usda.gov/soils/monitoring.shtml>
- Forest Service Soil-Disturbance Field Guide:
https://www.fs.usda.gov/t-d/php/library_card.php?p_num=0819+1815P
- NRCS Web Soil Survey and SSURGO:
<https://www.nrcs.usda.gov/resources/data-and-reports/web-soil-survey>
- Forest Service National Best Management Practices Program:
<https://www.fs.usda.gov/naturalresources/watershed/bmp.shtml>
- East Crazies example project intake assumptions: land exchange, proposed trail construction, proposed trail reroute, trailhead improvements, parcel restrictions, easements, wetland/riparian interfaces, grazing-related improvements, and acquired-parcel management needs.